

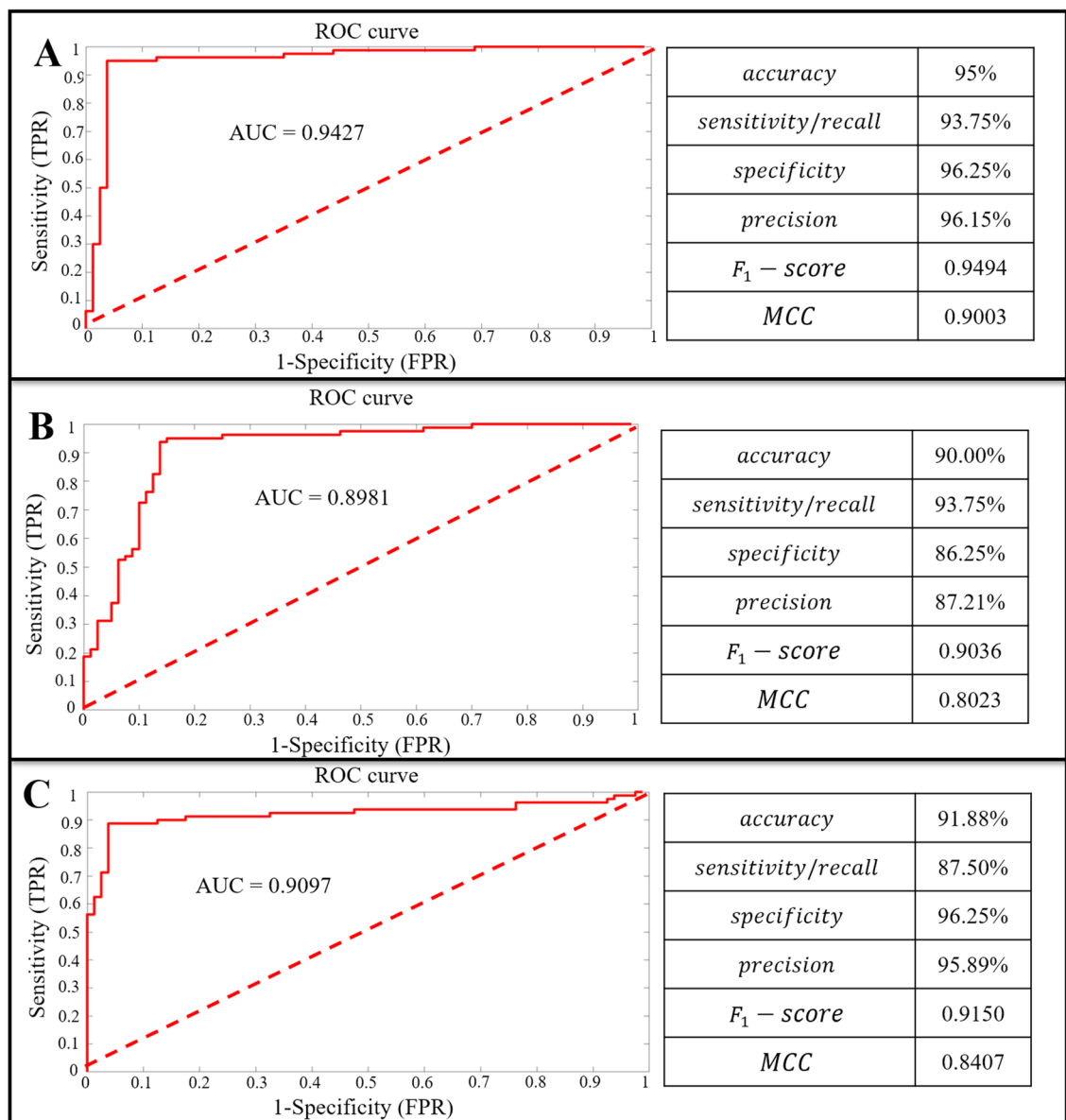


Figure 1. The classifier of the DNN models based on the matrices M , $M_{kinematics}$, $M_{kinetics}$. (A) The classifier of the DNN models based on the matrices M . (B) The classifier of the DNN models based on the matrices $M_{kinematics}$. (C) The classifier of the DNN models based on the matrices $M_{kinetics}$. ROC: Receiver Operating Characteristic; AUC: Area Under the ROC Curve; MCC: Matthews Correlation Coefficient; TPR: True Positive Rate; FPR: False Positive Rate.

1%-47% stance phase (contribution: 52.54%) was higher than the contribution of variables during the 48%-100% stance phase (contribution: 47.46%) to the successful classification.

The summed contribution of the relevance score of each joint (ankle, knee, hip) of each plane (sagittal, frontal, transverse) of kinematics (joint angle) and kinetics (joint moment) trajectories are showed in Fig. 2C. The summed contribution rate of the relevance score of the ankle, knee, hip was 43.16%, 35.98%, 20.86%, respectively. The summed contribution rate of the relevance score of the sagittal, frontal, transverse was 39.90%, 32.24%, 27.86%, respectively. The most relevant trajectory variables were the ankle dorsiflexion-plantarflexion angle (9.69%), the knee internal-external rotation angle (9.59%), the ankle dorsiflexion-plantarflexion moment (9.37%), and the knee flexion-extension moment (9.39%). Secondly, the relevant trajectory variables were the knee flexion-extension angle (7.19%), the hip abduction-adduction angle (8.64%), and the ankle inversion-eversion moment (7.93%). However, there was little relevance score in the variables of knee abduction-adduction angle (1.93%), hip flexion-extension angle (1.90%), hip internal-external rotation angle (1.85%), ankle internal-external rotation moment (2.99%), knee abduction-adduction moment (1.70%), hip flexion-extension moment (2.36%), hip internal-external rotation moment (1.18%).

The detailed distribution of relevance score during each joint (ankle, knee, hip) of each plane (sagittal, frontal, transverse) of kinematics (joint angle) and kinetics (joint moment) are showed in Fig. 2B. There were revealing